

View Reviews

Paper ID

45

Paper Title

Cost-Asymmetric Physics-Informed Forecasting for Robust NHS Critical-Care Surge Allocation

Reviewer #2

Questions

1. Overall Recommendation

Accept

2. Confidence

Confident

3. Relevance to the UK workshop on computational intelligence

Highly relevant

4. Strengths

- + the paper addresses a highly relevant and operationally important problem: coupling regional critical-care demand forecasting with surge-bed allocation under uncertainty.
- + the integration of physics-informed learning, quantile forecasting, online recalibration, and mathematical optimization is well aligned with the workshop's computational-intelligence scope.
- + the manuscript is technically substantial and generally transparent.
- + the experimental design includes several useful baselines, a chronological regime-shift evaluation, model ablations, rolling-origin testing, and a statistical comparison with ARIMA. Table 1 on page 9 shows that the recalibrated model performs strongly at 7-, 14-, and 21-day horizons and remains competitive at 28 days.
- + the optimization component is clearly formulated.
- + the sensitivity analysis is particularly valuable as the paper does not hide the fact that, with unconstrained transfers, most budget-exhausting policies tie on coverage and the main benefit of optimization is reduced transfer burden. It then demonstrates that allocation placement becomes more important when the travel network or transfer service

is constrained.

+ Fig.1 provides a clear overview of the forecast-to-decision pipeline, while Fig.2 and 3 help explain the epidemiological model and allocation behavior.

5. Weaknesses

- there are concerns regarding the description of the training loss as cost-asymmetric. The $q=0.9$ pinball component does penalize under-prediction nine times more than over-prediction. However, the complete loss gives equal weight to $q=0.1$, $q=0.5$, and $q=0.9$. The $q=0.1$ branch has the opposite asymmetry, so the aggregate three-quantile objective is symmetric when the quantiles are equally weighted. The upper quantile is cost-asymmetric, but the total training loss is not clearly biased against under-prediction.

- the paper does not report direct quantile-forecast evaluation. Since the allocation model is driven by the $q=0.1$, $q=0.5$, and $q=0.9$ outputs, the manuscript should report empirical interval coverage, pinball loss by quantile, weighted interval score, calibration by horizon, sharpness, and quantile-crossing frequency. Point-forecast RMSE and MASE do not establish that the scenario forecasts used by the optimizer are calibrated.

- the recalibration procedure improves the median point forecast but deliberately leaves the 80% interval unchanged. Consequently, the strongest forecasting improvement reported in Table 1 does not directly improve the uncertainty scenarios used by the allocation model.

- the scenario construction is highly stylized. Setting every region and horizon simultaneously to its $q=0.9$ value creates a co-monotone high-demand scenario. The authors acknowledge that this is conservative, but the resulting scenario probabilities (0.2,0.6,0.2) are representative weights rather than calibrated probabilities. Therefore, quantities labelled “expected unmet demand” are expectations under a designed scenario mixture, not necessarily estimates of real-world expected shortage.

6. Comments to Authors

The combination of physics-informed forecasting and robust regional allocation is relevant, technically interesting, and presented with more methodological transparency than many forecast-to-decision studies. The acknowledgement of identifiability limitations, the common LP evaluation of all policies, and the discussion of the unconstrained-transfer coverage tie are especially appreciated.

The most important issue is the interpretation of the multi-quantile objective. Eq.4 gives equal weight to $q=0.1$, $q=0.5$, and $q=0.9$. Although the $q=0.9$ term penalizes under-prediction nine-to-one, the $q=0.1$ term penalizes over-prediction by the same ratio. Summed with equal weights, the overall loss has balanced aggregate penalties for positive

and negative errors.

The manuscript should therefore either describe the method as multi-quantile forecasting with a cost-relevant upper quantile rather than an overall cost-asymmetric loss; assign greater weight to the upper-quantile branch; or introduce an explicit decision-weighted or asymmetric objective and compare it with the current formulation.

The quantile forecasts also require direct evaluation. Please report, separately for each horizon:

- empirical coverage of the nominal 80% interval;
- average interval width or sharpness;
- pinball loss for each quantile;
- weighted interval score or an equivalent proper scoring rule;
- under-prediction frequency for $q=0.9$; and
- the frequency and handling of quantile crossing.

Please also clarify the forecast pipeline after recalibration. Section 2.5 states that the median is recalibrated and blended with persistence, while the interval remains unchanged. If the allocation uses the original quantile heads, then the recalibrated RMSE improvements are largely separate from the allocation results. A diagram or explicit notation distinguishing the point forecast used for Table 1 from the scenario forecasts used by the LP would help.

The current “expected unmet demand” should be described carefully because it is calculated under representative scenario weights rather than calibrated scenario probabilities.

The paper should also discuss how much the results depend on the assumed baseline capacity, surge budget, and occupancy-to-capacity approximation.

The ablation results are useful, particularly the large degradation after removing the four PINN parameter features. However, because these parameters are not identifiable, I suggest avoiding the statement that the result proves the “physics prior” is responsible. A more defensible interpretation is that the structured PINN-derived representation adds predictive signal. An additional control using a similarly sized unconstrained time-feature network would help separate physical structure from representation capacity.

Questions

1. Overall Recommendation

Weak Accept

2. Confidence

Moderately confident

3. Relevance to the UK workshop on computational intelligence

Relevant

4. Strengths

The paper presents a well-motivated end-to-end framework that combines physics-informed forecasting with robust optimization for healthcare resource allocation.

The integration of quantile forecasting and robust LP optimization is novel and operationally meaningful.

The forecasting methodology is carefully designed, including asymmetric pinball loss, recalibration, and informative ablation studies.

Experimental comparisons against several forecasting baselines are comprehensive, and statistical significance is reported for key results.

The optimization model is clearly formulated and computationally efficient.

5. Weaknesses

The optimization evaluation is relatively limited. Most experiments are based on a single NHS case study and, in several settings, different allocation methods achieve identical coverage, making it difficult to assess the practical advantage of the proposed robust LP. Several planning parameters (budget, travel limits, transfer caps, scenario weights) appear to be selected heuristically, with limited justification or discussion of their real-world validity. The manuscript is quite dense and occasionally difficult to follow, particularly in Sections 2 and 3 where many model components are introduced simultaneously.

6. Comments to Authors

The paper addresses an important healthcare planning problem and presents an interesting integration of physics-informed forecasting with robust optimization. The forecasting component is particularly strong and appears to provide meaningful improvements over competitive baselines.

Although the LP formulation is elegant, many experiments show similar coverage across competing methods, making the practical benefit of the proposed allocation strategy less convincing. Additional experiments under more diverse demand scenarios, alternative datasets, or more realistic uncertainty models would strengthen the paper considerably. It would also be helpful to provide further justification for the selected planning parameters and scenario probabilities, or discuss how these could be estimated in practice. Finally, future research directions should be mentioned in the Conclusions.